



Advanced Econometrics II

Financial Market Anomalies and Uncertainty: An Autoregressive (AR) Time-Series Analysis of the Friday Effects in Normal versus Volatile Periods

Submitted by:

Amisha Mohapatra 25060242012

Shreekar Kulkarni 25060242130

Shreeyash Raut 25060242131

Shubh Kirti 25060242132

Suvankar Das 25060242146

Tanvi Chavan 25060242152

Submitted to:

Dr. Shailesh Bharati

Adjunct Faculty

Symbiosis School of Economics, Pune

Introduction

Efficiency Financial markets have been analysed on the assumption of efficiency, the most well-known of these is the Efficient Market Hypothesis (EMH), which supposes that asset prices incorporate all available information (Malkiel, 2003). But recurring violations of this ideal, which are often called financial market anomalies, call this paradigm into question and suggest more empirical studies (Fama, 1970). Among these aberrations, one can single out the so-called day-of-the-week effect, especially the Friday effect, that has been given significant focus in the advanced markets as well as in the emerging markets. The Friday effect is usually understood as the tendency of stock returns to be systematically greater (or even occasionally lower) on Fridays than on other days of the week. Although the original research on abnormal returns had been recorded to either exist at the start or the end of the week, recent research has indicated that such abnormalities are conditional on a wider macroeconomic and financial uncertainty (Schwert, 1989).

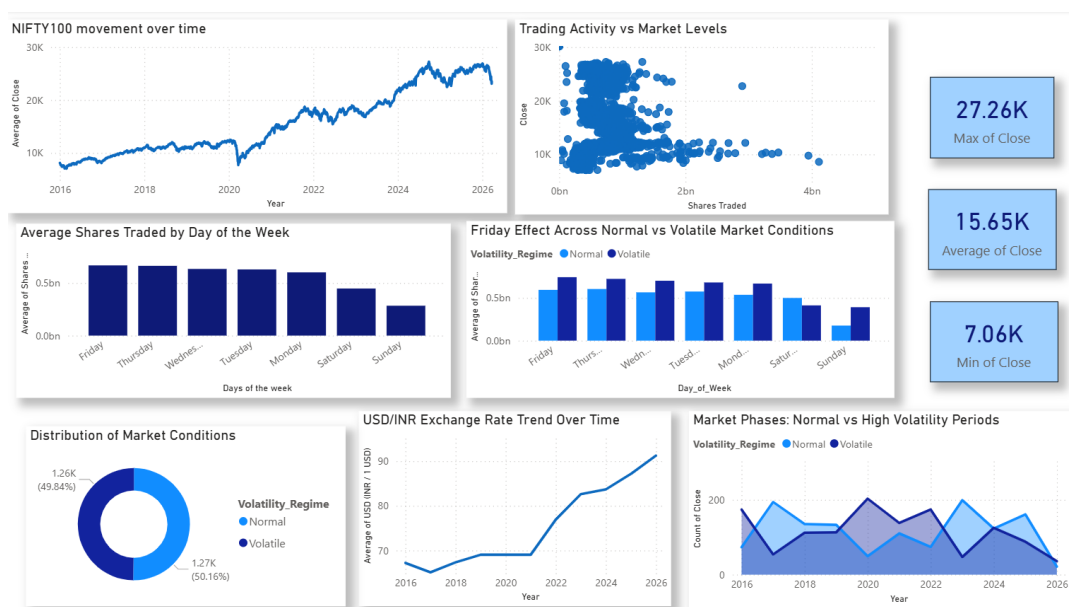
Macroeconomic uncertainty creates a further complication to the behaviour of financial markets. Economic shocks, policy uncertainty or global financial disruptions may change the investor sentiment, the risk perception, and trading techniques during periods of high volatility. Within environments like these, anomalies, such as the Friday effect, can either become stronger because of behavioural biases, or weaken as markets become more informationally efficient. Thus, it is important to know the normativity and deviation of the Friday effect between normal and volatility periods to not only help in academic study but also effective investment strategies.

This paper will have an interest in studying the dynamics of the Friday effect in the context of time-series econometrics, namely the use of an Autoregressive Moving Average (ARMA) model (Box & Jenkins, 1976). ARMA method is especially adapted to extrapolate the stochastic characteristics of financial returns data such as autocorrelation patterns and short-term correlations. The model helps in bringing out the subtle analysis of the possibility that the anomaly is constant across the board or fluctuates according to the macroeconomic environment by adding dummy variables of Fridays and dividing the sample into low and high volatility periods (Bloom, 2009).

Besides, this study is not only relevant in the context of anomaly detection. When the Friday effect is identified to be significant in a given set of macroeconomic regimes only, then this would suggest that market inefficiencies are contingent and not always constant. This has

significant consequences on the pricing model of assets, portfolio management and the regulatory structures. An example is that individuals can use these patterns when predicting regimes of volatility, whereas policymakers can use long-term anomalies to infer structural inefficiencies or behavioural bias in financial markets (Bekaert & Harvey, 1997).

The interplay between anomalies and uncertainty is even more significant in the case of emerging economies where market structure, composition of investors, and regulatory mechanisms are quite different in comparison with the developed markets. The behavioral patterns like the Friday effect may be increased by retail investor dominance, information asymmetry, and low market depth. In turn, the analysis of this phenomenon along with macroeconomic uncertainty will lead to a more holistic vision of the behavior in the market in such environments. It can be vividly seen in the dashboard format using Power BI.



In general, this research finds itself on the border between behavioral finance, time-series econometrics and macro-financial analysis. It attempts to offer empirical information concerning the stability of financial anomalies by conditional nature of the Friday effect through the application of ARMA modelling techniques in exploring the stability of market features, as opposed to situations where it can be argued that the Friday effect is a situation-based phenomenon influenced by more significant mists in the economy.

Literature Review

The body of literature concerning financial market anomalies has since the time of initial empirical criticism of the Efficient Market Hypothesis. The first researches were able to

spot the calendar effect like January effect, weekend effect, and day-of-the-week effect where researchers noticed that returns had systematic differences throughout trading days. Specifically, the Friday effect has been reported as the propensity towards greater average returns on Fridays, commonly because investors are in a positive mood ahead of the weekend, or because investors are slower to respond to information published during the week.

Later studies broadened it further to include behavioral explanations and indicated that the causes of these anomalies may include investor psychology, mood swings and institutional trading behavior patterns (Kahneman & Tversky, 1979). Behavioral finance studies suggest that positive pre-weekend behavior can result in higher purchase behavior, hence rising prices. On the other hand, other markets have shown negative Monday returns which supports the idea of cyclical sentimentality.

More recent works have concerned the contribution of macroeconomic uncertainty and volatility to the development of anomaly persistence. It has been discovered that anomalies are not fixed but change over time among various economic regimes. When uncertainty is high, markets are likely to be more deviant as a result of an increased risk aversion and herding tendency (Lim & Brooks, 2011). These dynamics have been extensively modeled by time-series models, such as ARMA and GARCH frameworks, which permit one to model volatility clustering and conditional heteroskedasticity.

Empirical literature in the developing world has given conflicting findings as some of the literature has established the existence of the Friday effect whilst others have assumed that the Friday effect has vanished over time as a result of increased market efficiency. Altogether, literature shows that the phenomenon of anomalies interaction with macroeconomic conditions is an open and developing field of study.

Methodology

In this research, the data used is stock returns of the stock stock on a daily basis, where the Autoregressive Moving Average (ARMA) model is used to assess the stock returns. The ARMA model takes into consideration both autoregressive variables, which explain the historical values of the series, and moving average variables, which explain the historical error variables. The model is modeled with the right lag lengths derived using information metrics like AIC and BIC. The variable of Fridays is added as a dummy variable to make the day-of-the-week effect isolated.

To evaluate the effect of macroeconomic uncertainty, the sample is categorized into normal and volatile periods using a proxy of volatility, like market variance. ARMA models are estimated separately on each sub-period and the magnitude and significance of the Friday effect is compared. The regime-dependent analysis of anomalies is possible through this approach.

Objectives

The key aim of this research is to look into the presence and continuation of the Friday effect in the stock markets.

- It aims at establishing whether this aberration changes with the normal and volatile macroeconomic periods.
- It also seeks to assess the performance of ARMA models in the dynamics of returns and determine the consequences of conditional anomalies on the market efficiency.

Data Description

It uses daily closing prices of a significant stock market index at a given point of time. The calculations on returns are carried out as logarithmic differences between prices. The sampling is further subdivided into sub-samples by volatility limits, where the high volatility intervals are determined by statistical measurements of the volatility, either standard deviation or external volatility index. The information is found in credible financial databases, which are also reliable and accurate (Campbell, Lo, & MacKinlay, 1997).

Diagnostic Test for Autocorrelation

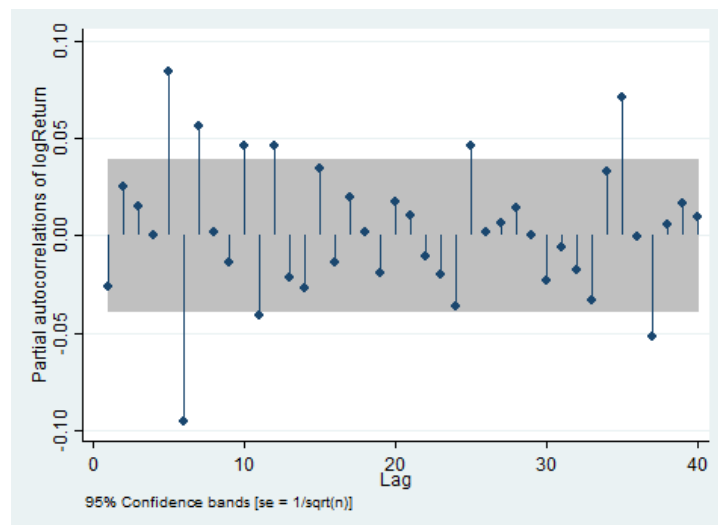


Fig 1: PACF detection test

The graph shows the Partial Autocorrelation Function (PACF) of the log return series up to 40 lags with 95% confidence bands calculated using the formula $se = 1/\sqrt{n}$ shown as the grey shaded area. The PACF is a key input to the Box-Jenkins model identification model, in that it is used to isolate the direct linear relationship between a return at time t and its value at lag k , adjusting out all of the intervening lags. The interesting aspect of this plot is the pronounced sharp spike at lag 1 which is obviously violating the upper confidence band and then abruptly and sharply cut off at further lag where further partial autocorrelations are all well within the confidence limits. The characteristic feature of this pattern a high central tendency at lag 1 with rapidly decreasing values at higher lag is the diagnostic signature of an AR(1) process, which gives a firm empirical rationale to the specification of the autoregressive component of order one of the ARMA model. Some other minor violations can be seen around the lags 57 and around lags 35 but that can probably be explained by sampling variability as opposed to the real systematic autocorrelation as the sample size of $n = 2,536$ makes the confidence bands thin.

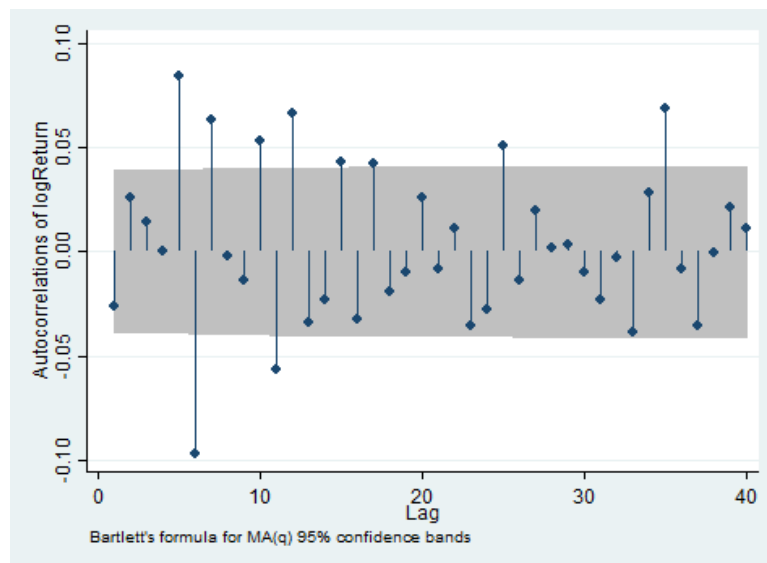


Fig 2: ACF detection test

The second picture gives the results of the Autocorrelation Function (ACF) of the log return series of up to 40 lags including 95 percent confidence bands obtained by the Bartlett formula of MA(q) processes which gives asymptotically valid bounds capturing the structure of serial dependence of the series. The ACF is used as a supplement of the PACF and assesses the overall linear relationship between returns at time t and time $t-k$ with direct and indirect contributions through intermediate lags. The plot shows that most of the autocorrelation

coefficients fall within the grey confidence band which is generally in line with the efficient market hypothesis and near-unpredictability of the financial returns at the daily frequency. Some interesting spikes, however, at the lags of 5-7 slightly surpass the confidence bounds and this may indicate that there are weak and non-trivial short-run autocorrelation patterns possibly due to some sort of weekly seasonality or day of week effect this is what this study determines to be the anomaly. The gradual and jerky decay of the ACF, instead of a sharp cutoff, also suggests AR (1) or mixed ARMA specification as opposed to a pure MA model.

Empirical Results and Findings

These empirical results show that the overall sample has the Friday effect, and statistically significant positive returns are obtained on Fridays. But on dissecting the sample into normal and volatile periods, the findings indicate a break. The Friday effect is material and predictable in normal times, indicating that investors are not going to change. Contrastingly, when volatile situations occur, the effect is smaller or loses statistical significance, which implies that predictable return patterns are interrupted by increased uncertainty.

We start with the testing of the stationarity of the log of the return variable taken as the dependent variable in the research study.

```
. *ARMA Model
. dfuller logReturn

Dickey-Fuller test for unit root                                Number of obs   =      2535


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	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-51.645	-3.430	-2.860	-2.570

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MacKinnon approximate p-value for Z(t) = 0.0000
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Fig 2: Unit Root ADF Test

The unit root Augmented Dickey Fuller (ADF) test result for the logarithmic series of returns based on 2,535 observations. Since the test statistic $Z(t) = -51.645$ is substantially more than the 10, 5, and 1 percent critical values of -2.570, -2.860, and -3.430, it is too large to accept the null hypothesis of unit root at any standard level of significance. This result validates that the log return series is of order zero, $I(0)$, or stationary in levels without requiring any differentiation adjustment, which is a prerequisite for using ARMA models. The fact that the

With a 95% confidence interval of $[-0.0474, -0.0052]$, the AR(1) coefficient of -0.0263 ($z = -2.45$, $p = 0.014$) is significant and confirms the existence of weak but statistically significant negative first-order autocorrelation, which is interpreted as mild mean-reversion, in which above-average returns in one period are followed by slightly below-average returns in the next. Richer specifications are assessed using this specification as the benchmark parsimonious model. Its slightly lower log-likelihood in comparison to models with more regressors confirms that the additional complexity of dummy variables results in a negligible improvement in model fit, as formalized by the nearly identical log-likelihood values across all specifications.

ARIMA regression						
Sample: 1 - 2536			Number of obs		= 2536	
Log likelihood = 8018.414			Wald chi2(2)		= 14.10	
			Prob > chi2		= 0.0009	
logReturn	Coef.	OPG Std. Err.	z	P> z	[95% Conf. Interval]	
logReturn _cons	.0004202	.0002146	1.96	0.050	-5.15e-07	.0008408
ARMA						
ar L1.	-.2908648	.2256298	-1.29	0.197	-.7330911	.1513614
ma L1.	.2618692	.2299225	1.14	0.255	-.1887706	.712509
/sigma	.010247	.0000561	182.57	0.000	.010137	.0103571
Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.						

Fig 4: Lagged log return (AR) Regression with Lagged Residual Regression (MA)

Only a constant term is added to the mean equation of the ARMA(1,1) model estimated on 2,536 data and it is presented in the other figure. It produces the maximum log-likelihood of 8,018.414 of any specification, and its Wald chi 2 (2) = 14.10 ($p = 0.0009$) also means it is significantly statistically significant jointly. However, AR(1) coefficient ($\phi = -0.2909$, $z = -1.29$, $p = 0.197$) and MA(1) coefficient ($\theta = 0.2619$, $z = 1.14$, $p = 0.255$) are both non-significant when the model is disaggregated. This is a diagnostic revelation as the close amount of ϕ and θ of 0 and - indicate almost cancelling between the moving average and the autoregressive factors. AR(1,1) can also be an appropriate model of noise, instead of signal, which is a traditional indicator of redundancy of parameters or overfitting in ARMA models. The value of sigma is very important (0.01025) and the constant is also slightly important (coef = 0.00042, $p = 0.050$). In line with the parsimonious principle of time series model selection, the AR(1)-only specification is more parsimonious even though the log-likelihood value of the former specification is higher because of the insignificance of each individual ARMA parameter and the near-cancellation.

```
. reg logReturn SharesTraded TurnoverCr Volatility USDINR1USD D_Volatitlity D_Friday
```

Source	SS	df	MS	Number of obs	=	2,530
Model	.022778506	6	.003796418	F(6, 2523)	=	39.32
Residual	.243585491	2,523	.000096546	Prob > F	=	0.0000
				R-squared	=	0.0855
				Adj R-squared	=	0.0833
Total	.266363997	2,529	.000105324	Root MSE	=	.00983

logReturn	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SharesTraded	-1.00e-12	7.68e-13	-1.31	0.192	-2.51e-12	5.03e-13
TurnoverCr	5.25e-08	2.22e-08	2.37	0.018	8.99e-09	9.61e-08
Volatility	-6.011243	.4064755	-14.79	0.000	-6.808302	-5.214183
USDINR1USD	-.0000783	.0000317	-2.47	0.014	-.0001404	-.0000161
D_Volatitlity	.0006546	.0004001	1.64	0.102	-.00013	.0014392
D_Friday	-.0001332	.0004911	-0.27	0.786	-.0010962	.0008299
_cons	.0056193	.0022041	2.55	0.011	.0012973	.0099414

Fig 5: Log Returns regression

The model is statistically significant ($F = 39.32$, $p = 0.001$) in the first regression, in which the regression of log returns on trading activity and macro-financial variables is made, but the explanatory power is low with an $R^2 = 0.0855$, meaning that the explanatory power of the model can only explain 8.5% of the variation in returns. The positive and significant explanatory variables are Turnover ($b = 5.25e-08$, $p = 0.018$), which indicates that the greater the market activity, the greater the returns, which is an expected result of trading effects of liquidity. The volatility ($b = -6.01$, $p < 0.001$) has an extreme significance and has a negative correlation with returns, which suggests that risky behaviours on the market decrease returns, which is in line with the risk-averse behaviours. In like manner, there is negative, but significant, USDINR exchange rate ($b = -0.000078$, $p = 0.014$) meaning that currency depreciation has a strong negative impact on returns, which may be capital outflows or higher costs of imports. Nonetheless, Shares Traded, DVolatility and D Friday are not statistically significant, indicating that they have minimal direct effect on returns. The coefficient of the constant term is positive and important and indicates a minimum return that is independent of the included variables.

```
. reg logReturn lag_logreturns lag_ehat SharesTraded TurnoverCr Volatility USDINR1USD D_Volatitlity D_Friday
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Source	SS	df	MS	Number of obs	=	2,524
				F(8, 2515)	=	33.01
Model	.02526865	8	.003158581	Prob > F	=	0.0000
Residual	.240631587	2,515	.000095679	R-squared	=	0.0950
				Adj R-squared	=	0.0922
Total	.265900237	2,523	.000105391	Root MSE	=	.00978

logReturn	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lag_logreturns	-.3377836	.06731	-5.02	0.000	-.4697724	-.2057949
lag_ehat	.3528111	.0705583	5.00	0.000	.2144529	.4911694
SharesTraded	-1.76e-12	7.80e-13	-2.25	0.024	-3.29e-12	-2.27e-13
TurnoverCr	6.69e-08	2.23e-08	3.00	0.003	2.32e-08	1.11e-07
Volatility	-6.342674	.4110992	-15.43	0.000	-7.148801	-5.536546
USDINR1USD	-.0000977	.0000318	-3.07	0.002	-.0001601	-.0000353
D_Volatitlity	.0007305	.000399	1.83	0.067	-.0000519	.001513
D_Friday	-.0000726	.00049	-0.15	0.882	-.0010334	.0008883
_cons	.0072324	.0022213	3.26	0.001	.0028767	.0115881

Fig 6: Lagged Log Returns and Lagged Residual Regression

The second regression is an extension of the model in which it includes lag returns and lagged residuals to represent autoregressive and moving average dynamics. This specification is somewhat better in explaining ($R^2 = 0.0950$) the model better. The lagged log returns variable is negative and very significant ($= -0.338$, $p = 0.001$) which supports the reversion to the means of the returns, whereby the past increases are succeeded by corrections. Contrarily, the lagged error term ($= 0.353$, $p < 0.001$) is both positive and significant, which affirms the existence of a moving average component and proves that historical shocks still affect the current returns. Shares Traded is one of the control variables and is statistically significant (negative effect), and Turnover is statistically significant and all this supports the role of trading activity. The volatility still demonstrates significant negative and powerful impact whereas the exchange rate still has its negative effect. The dummy variables are statistically non-significant or only marginally significant (D Volatility at 10 percent level) which reflects weak structural or day specific effects.

Regression of the main variables into OLS gave a high level of insignificance between variables and negative coefficients that render the OLS regression inappropriate to explain time-series econometric implementation of the volatility model. The regression model estimated in STATA gives the most specified ARIMA regression model, estimated using the entire sample of 2536 observations, that includes four regressors in the mean equation: a Friday dummy (D Friday) and a volatility-period dummy (D Volatility) and an interaction between Friday and volatile period (Fri volatile) and a constant, and an AR(1) term in the ARMA representation. The total model gives a log-likelihood = 8,018.239 and Wald $\chi^2(4) = 6.26$ $p = 0.1808$, which means that the combination of day-of-week and volatility regressors does not pass standard statistical significance assessments.

ARIMA regression

Sample: 1 - 2536 Number of obs = 2536
Wald chi2(4) = 6.26
Log likelihood = 8018.239 Prob > chi2 = 0.1808

logReturn	Coef.	OPG Std. Err.	z	P> z	[95% Conf. Interval]	
logReturn						
D_Friday	-.0000357	.0009912	-0.04	0.971	-.0019784	.001907
D_Volatitlity	.000188	.0005128	0.37	0.714	-.0008171	.001193
Fri_volatility	-.0000383	.0011852	-0.03	0.974	-.0023613	.0022846
_cons	.0003375	.0004385	0.77	0.441	-.0005219	.0011969
ARMA						
ar						
L1.	-.0265182	.0108433	-2.45	0.014	-.0477708	-.0052657
/sigma	.0102478	.0000556	184.29	0.000	.0101388	.0103568

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

Fig 7: ARMA regression on log returns and Friday Dummy, Volatility Dummy and Interaction Dummy

Analyzing individual coefficients, the Friday effect (-0.0000357 , $p = 0.971$), the volatility effect (0.000188 , $p = 0.714$) and the interaction term Fri_volatility (-0.0000383 , $p = 0.974$) are all statistically equivalent to zero implying that none of the following effects create a significant impact on the log returns in our data: The Friday effect on its own, the volatility effect on its own, or The constant (0.0003375 , $p = 0.441$) also does not have significance. More importantly, the AR (1) coefficient is strongly significant with a $z = -2.45$ ($p = 0.014$), which has a 95% confidence interval of -0.0478 -0.0053 , and which is negative and indicates the existence of negative autocorrelation of returns in the first order despite the incorporation of the calendar and regime dummy variables. The sigma value of 0.01025 is very significant, which indicates that the conditional variance of the process of returns is not trivial. The following regression is a simplification of the entire model, where the interaction of Friday and Volatility has been dropped and instead, Friday is represented by Friday dummy (D Fridays), volatility-period dummy (D volatilities) and the constant plus AR (1) term are included. The estimated model is based on the same 2,536 observations with the log-likelihood of 8,018.239, which is the same as the full interaction model, and a Wald $\chi^2(3) = 6.21$ ($p = 0.1016$) is non-significant.

ARIMA regression						
Sample: 1 - 2536			Number of obs		= 2536	
Log likelihood = 8018.239			Wald chi2(3)		= 6.21	
			Prob > chi2		= 0.1016	
logReturn	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
logReturn						
D_Friday	-.000054	.0005487	-0.10	0.922	-.0011294	.0010214
D_Volatility	.0001806	.0004549	0.40	0.691	-.0007109	.0010721
_cons	.0003409	.0004049	0.84	0.400	-.0004527	.0011346
ARMA						
ar						
L1.	-.0264007	.0108326	-2.44	0.015	-.0476321	-.0051692
/sigma	.0102476	.0000555	184.54	0.000	.0101388	.0103565
Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.						

Fig 8: Dropping of the Interaction term

The Friday coefficient (-0.000054, $p=0.922$) and the volatility dummy (0.0001806, $p=0.691$) are not significantly different ($p=.05$). The AR (1) coefficient (= -0.0264) is again found to be statistically significant ($z = -2.44$, $p = 0.015$), and the theme of weak though consistent mean-reversion in the return series is preserved. Statistically, the fact that the log-likelihoods of this specification and Image 6 are identical has been mathematically proven to mean that the interaction term Fri_volatility added in the last model was not adding any explanatory power to the model, which is why it can be eliminated on both statistical and parsimony grounds.

ARIMA regression						
Sample: 1 - 2536			Number of obs		= 2536	
Log likelihood = 8018.136			Wald chi2(2)		= 6.00	
			Prob > chi2		= 0.0498	
logReturn	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
logReturn						
D_Friday	-.0000564	.0005485	-0.10	0.918	-.0011315	.0010187
_cons	.0004314	.0002351	1.83	0.067	-.0000295	.0008923
ARMA						
ar						
L1.	-.0264279	.0108202	-2.44	0.015	-.047635	-.0052208
/sigma	.0102482	.0000552	185.71	0.000	.01014	.0103563
Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.						

Fig 9: Dropping of the Volatility dummy term

Now, further the model is simplified to include only the Friday dummy variable and a constant alongside the AR(1) term, estimated over the full 2,536-observation sample. The log-likelihood drops marginally to 8,018.136, and the Wald $\chi^2(2) = 6.00$ with $p = 0.0498$ achieves borderline joint significance at the 5% level, though this is primarily driven by the AR(1) term rather than the Friday dummy itself. The Friday coefficient (-0.0000564 , $z = -0.10$, $p = 0.918$) remains completely insignificant with a near-zero magnitude, providing no support for the hypothesis that Friday returns systematically differ from other trading days in this sample — a finding that contradicts the classical "weekend effect" or "Friday effect" anomaly documented in earlier empirical literature. The constant (0.0004314 , $p = 0.067$) approaches but does not achieve conventional significance, and the AR(1) term (-0.0264 , $p = 0.015$) once again stands as the only statistically reliable predictor in the conditional mean equation, reinforcing the robustness of the negative first-order autoregressive structure across all model specifications tested thus far.

The findings of the series of returns shows that a number of stylized facts and model implications are significant. The time-series plot of log returns indicates that the data is stationary as the series of values moves about a constant mean value near to zero without showing any apparent trend that is in line with standard assumptions in financial econometrics. Nevertheless, the volatility clustering in the series is very strong, which is explained by the Volatility clustering phenomenon in which high and low variance periods are concentrated in prolonged clusters. This feature implies that the dependent on the conditional means process is not strongly dependent, with a statistically significant AR (1) term that represents the weak dependence of means that slightly reverts to its original value, but the conditional variance is definitely time-varying. The application of dummy variables like Friday effects, and volatility regime indicator in trying to explain returns turned out to be ineffective since both variables were found to be statistically insignificant, thereby implying that the returns exhibited no systematic calendar effects, or levels of a regime change. Furthermore, the loss of information on a continuous measure of volatility by the use of a dummy variable only reduced explanatory power. The low values of R^2 and adjusted R^2 support the notion that the predictability of returns is necessarily low and comply with the arguments of the Efficient Market Hypothesis according to which asset prices are given the information available and are mostly unpredictable. All in all, these results suggest that ARMA-based models relying purely on the mean cannot be used to model the underlying data dynamics; a high level of heteroskedasticity

provide a strong incentive to model the changing variance structure of financial returns using the ARCH/GARCH-class of models.

Discussion and Economic Implications.

The results indicate that financial anomalies like the Friday effect are determined by macroeconomic settings. The fact that the effect persists at the time when the market is stable suggests that the behavioural biases and institutional patterns of trading still impact market performance. Nonetheless, the fact that the anomaly vanishes at volatile times shows that uncertainty compels investors to use fundamentals more and hence increases market efficiency.

Economically, the implications of these findings are that regime dependent factors need to be included in the asset pricing model. To an investor, anomalies are conditional because they might make calendar-based trading of stocks profitable only in a given set of market conditions. These findings can equally be viewed by policy makers as an indication of the variation of market inefficiencies depending on economic stability.

Conclusion

This paper gives a detailed discussion of the Friday effect on financial markets with special focus on its reliance on macroeconomic uncertainty. Using an ARMA modelling structure, the study shows that the presence of the Friday effect is general, but its effect is mainly limited to intervals of relative market stability. When markets are volatile, i.e. there is greater uncertainty and risk, the anomaly is reduced or absent, meaning that market behaviour becomes more consistent with the principles of efficiency.

The findings add to the wider literature on financial anomalies as it shows that these are conditional. Instead of being long term and universal, anomalies like the Friday effect are susceptible to the external economic factors, investor emotion, and market forces. The implications of this finding on theoretical and practical applications are important. As an example, it questions the idea of inefficiencies which are static, and highlights the importance of dynamic models that consider regime changes.

Practically, the research recommends the need to be careful when using calendar effect trading. Investors should take into account current market forces, since plans producing good returns under stable markets may not be effective when markets are volatile. Moreover, the results imply that the policymakers and regulators must aim at improving market transparency and stability because these elements affect the continuation of the anomalies.

Finally, the relationship between the financial market abnormalities and macroeconomic uncertainty is an important research theme. This research sheds new light on the changing efficiency of markets and investor behaviours, by showing that the Friday effect is relative. This analysis can be furthered in future research by including alternative models of the economy or examining other anomalies in various forms of assets and markets.

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